

Cathy Fang

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Education

National University of Singapore

Master of Science in AI for Science; GPA: 4.67 / 5.00

Aug 2025 – Jan 2027

Singapore

Beijing University of Chemical Technology

Bachelor of Engineering in Computer Science; GPA: 90.67 / 100

Sep 2021 – July 2025

Beijing, China

Experience

Micron Technology

ML/AI Engineer Intern

Aug 2026 – Present

Singapore

• Agent Skill Quality Evaluation and Semantic Deduplication

- **Quality evaluation:** Built automated pre-release evaluation covering approximately **400 skills**, integrating format validation, security checks, and privacy detection; produced severity-ranked findings to support skill review and correction.
- **Semantic deduplication:** Used embedding similarity to detect repeated instructions within skills and capability overlap across skills; applied LLM verification to distinguish internal redundancy from necessary supporting explanations, supporting context reduction and skill-library maintenance.

Tencent (Interactive Entertainment Group, IEG)

Research Intern

May 2026 – Jul 2026

Singapore

• Chemical Reasoning Model: Online Self-Evolution Training

- Built an “online sampling–answer verification–feedback distillation–policy update” loop with verl and Qwen3-8B on 8×A800 GPUs; turned successful same-question trajectories into token-level signals, raising SciKnowEval chemistry reasoning accuracy averaged over 16 samples from **40.7% to 79.1%**.

• Multi-Turn Data Science Agent: Online Post-Training

- Built a Kaggle-style multi-turn Agent pipeline from task analysis and Python execution to feedback correction and submission; generated trajectories online with the current policy, used sandbox results and task scores as verifiable feedback, and converted long trajectories into turn-level distillation samples.
- Constructed same-task multi-trajectory feedback from stronger trajectories, runtime errors, and intermediate scores; added tool-call retries and token-budget controls, ran **20+** training and ablation experiments, and identified reduced tool use and output homogenization late in training.

Microsoft (Data, Knowledge, Intelligence, DKI)

Research Intern

Dec 2023 – Oct 2024

Beijing, China

• SheetMontage: Multi-Task SFT Data Synthesis and Model Training for Excel Agents

- **Corpus construction:** Built format cleaning, language filtering, and MinHash-LSH deduplication, reducing **560K+** raw worksheets to **290K+** high-quality real-world spreadsheets; extracted formulas and cell ranges for task construction.
- **Data synthesis:** Synthesized instructions for cell-range grounding, formula generation, and chart creation from worksheet context and target operations; generated multiple queries per target, filtered them with an independent LLM for accuracy, clarity, and completeness, and manually reviewed evaluation samples.
- **Model training and evaluation:** Built task-specific benchmarks from real worksheets and Excel documentation, balancing frequency, diversity, and difficulty; compared few-shot, LoRA, and full SFT on LLaMA-3.1-8B, reaching **97%/71%/68%** accuracy across the three tasks.

• Excel Agent: File Understanding, NL2DSL Generation, and Reliability Optimization

- **File understanding:** Converted complex workbooks into compact, structured model context, aligning user intent with worksheet content and structure.
- **Structured generation:** Built intent-recognition and generation workflows that map requests to JSON with executable DSL; applied format and parser validation before Excel execution for analysis and visualization.
- **Reliability optimization:** Classified output errors and fed validation results into bounded retries, repairing **95%** of compilation errors within one retry; shipped globally.

• Multilingual Translation Pipeline

- Built an automated Excel translation pipeline supporting **26 languages** at approximately **90%** accuracy, saving **RMB 3M** in labor costs.

Volkswagen (New Energy PFME)

Research Intern

Jul 2023 – Aug 2023

Shanghai, China

- Calibrated vision-sensor data and standardized defect labeling; optimized Cognex VisionPro localization and defect segmentation rules to reduce false positives and negatives.
- Achieved **92%** FOD detection and **90%** adhesive continuity detection in on-site trials, reducing manual rechecks.

Projects

Sequence-Augmented GraphGPT for PPI Link Prediction

Research Assistant

Aug 2025 – Dec 2025

NUS × Alibaba

- **Data construction:** Filtered a human-specific subgraph with **19K proteins** from a multi-species protein association graph; aligned protein IDs, amino-acid sequences, and ESM-2 representations for **95.5%** of nodes, and constructed link-prediction training and evaluation sets.
- **Model training:** Used the GraphGPT graph Transformer to serialize local interaction networks around candidate protein pairs into graph tokens; integrated frozen **ESM-2** sequence representations through gated fusion and trained a PPI link-prediction model.
- **Evaluation:** Compared multiple graph-sequence fusion strategies with a graph-only baseline; gated fusion achieved **0.941 AUROC**, improving by **0.8 percentage points**, and was evaluated for cross-dataset generalization on external human PPI data.

Publications

Sole First Author | *Classification of Diabetic Retinopathy Based on Channel Attention and Hierarchical Multitasking*, Chinese Journal of Stereology & Image Analysis, 2023.

Skills

Languages: Chinese, English; **Programming:** Python, SQL, C++, C#, Java.

AI Stack: PyTorch, Transformers, verl; SFT/LoRA/RL, Tool Calling, Embedding Retrieval, LLM Judge.

Awards and Scholarships

Honors: Beijing Outstanding Graduate; Liwen Yangyan Scholarship (top **1%**); Outstanding Student (top **3%**); First-Class Scholarship; Outstanding League Member.

Competitions: CCPC Women’s Finals Bronze (2022–2023); RAICOM National First (2023); Computer Skills C++ Provincial First (2024); Lanqiao Cup C++ Provincial Second (2023); RoboCom Provincial Third (2022).